CareFlow: AI Care Coordination Assistant

Background

Navigating healthcare can be overwhelming for patients and administratively exhausting for care providers. CareFlow is designed to be an empathetic, intelligent digital assistant that bridges this gap. By handling the heavy lifting of care coordination—from gathering initial patient details to managing follow-ups—CareFlow allows healthcare professionals to focus on what matters most: patient care.

Problem Statement

Scope: Healthcare coordination is currently highly fragmented. Patients have to repeat their medical histories, care teams struggle to access up-to-date guidelines while interacting with patients, and administrative bottlenecks delay critical care.

Key Challenges:

- **Data Silos & Amnesia:** Systems lack persistent memory, forcing patients to repeat their stories.
- **Unstructured Chaos:** Intake processes are often messy, leading to missing crucial information.
- **Safety Risks:** Using standard AI in healthcare risks "hallucinations" or unsafe medical advice.
- **System Disconnect:** Existing tools (scheduling, EHRs) don't naturally talk to each other without complex integrations.

Core Deliverables & Technical Implementation

To solve these challenges, CareFlow will be built as an end-to-end, multi-agent AI system containing the following mandatory components:

1. **Structured Intake System:** An intelligent conversational interface that naturally but systematically collects required patient data (symptoms, history, demographics) without feeling like a rigid form.
2. **Persistent Memory:** A secure memory module so the AI remembers past interactions, treatment plans, and patient preferences across multiple sessions.
3. **Production RAG (Retrieval-Augmented Generation):** A secure knowledge-retrieval pipeline that grounds the AI's responses in verified medical guidelines, clinical documents, and hospital protocols.
4. **Multi-Agent Workflow Orchestration (LangGraph):** A robust state-machine using LangGraph to seamlessly coordinate different AI "agents" (e.g., an Intake Agent, a Triage Assistant, a Scheduling Agent).

5. **Tool Integration & MCP (Model Context Protocol):** Secure integration with external tools (like calendars, EHR databases, and communication platforms) using standard MCP for seamless data exchange.
6. **Observability & Reliability Monitoring:** Dashboards and logging mechanisms to track AI performance, trace decision-making paths, and ensure the system is reliably available.
7. **Safety Guardrails & End-to-End Evaluation:** Strict behavioral boundaries to prevent the AI from offering unverified diagnostic advice, coupled with an evaluation framework to continuously test accuracy and compliance.

Expected Output (Project Verification Criteria)

At the conclusion of this project, the following outcomes must be achieved and demonstrable for verification:

- **A Functional Multi-Agent Application:** A working prototype of CareFlow where a user can interact with the system from initial greeting through a complete intake process.
- **Demonstrable Memory & RAG:** The ability to show the AI accurately recalling a simulated patient's previous visit, and successfully pulling verified information from a provided document database.
- **Actionable Tool Use:** Proof of the AI successfully triggering external tools (e.g., booking a mock appointment or updating a structured database via MCP).
- **Safety Assurance:** Documentation and live testing proving that if asked for a medical diagnosis, the AI safely deflects and adheres to its guardrails.
- **Performance Metrics:** An observability report detailing the system's reliability, response times, and task completion rates during testing.
- **GuardrailGuarantee:** Show how the application constrain against the irrelevant domain question.

Intended Outcome: A streamlined, safe, and highly reliable AI care coordinator that reduces administrative burden on clinic staff by 30-40% while providing patients with a seamless, personalized, and continuously contextual care experience.